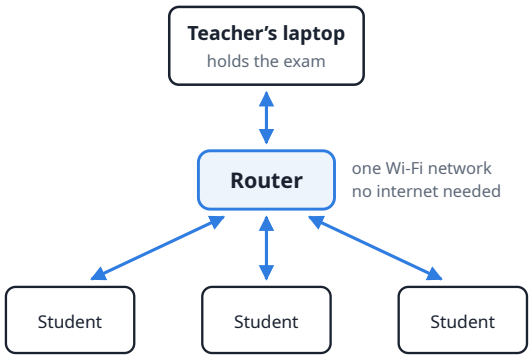
Classroom network — setting up the router

Preparing a Wi-Fi router for exams with the Virtual Optics Laboratory · version 1.3.0 or later

An exam runs on the teacher's laptop. The students' laptops connect to it through a Wi-Fi router in the room and fetch the questions, send their answers every few seconds and hand them in. **No internet is needed**: the router does not have to be connected to anything.

What the router **does** have to do is let the laptops talk to each other. Most home routers do this out of the box, but some settings — and phone hotspots, guest networks and laptops that drift onto another network — stop it. This guide sets the router up once and gives a check to run before each exam.



> **The three things that matter.** 1. Every laptop on the **same** Wi-Fi network of the router. 2. **AP isolation off** on that network. 3. The teacher's laptop installed with the **teacher's installer**, with its administrator prompt accepted, so that Windows Firewall lets the students in.

1. Choosing the router

- **Any home Wi-Fi router** will do. It needs power only; leave the internet (WAN) socket empty.
- **Use a router of its own for the exam room** rather than the building's network: school and university networks usually keep devices apart on purpose.
- **Check how many devices it accepts.** Some small routers stop at about 30 wireless devices. Count every student laptop plus the teacher's, and leave a few spare. The limit, if there is one, is in the router's manual or its wireless settings (*maximum clients*).
- **A phone hotspot** works for a few laptops (a test, a small group) but not for a class: phones accept only a handful of devices, and the hotspot disappears when the phone locks or its battery runs low.
- Put the router **in the middle of the room**, raised, away from metal cabinets.

2. Setting up the router (once)

1. **Open the router's settings page**

   Connect a laptop to the router's Wi-Fi (or with a network cable). Open a web browser and type the address printed on the label underneath the router — usually `192.168.0.1` or `192.168.1.1`; TP-Link routers also answer at `tplinkwifi.net`. Log in with the password on the label, or the one chosen the first time the router was set up. This works without internet.

2 Turn AP isolation off — the most important setting

AP isolation lets every device reach the internet but not each other. With it on, the students' laptops cannot see the classroom, and nothing on a laptop can fix that. Look in the wireless settings, often under **Wireless → Advanced**, for any of these names, and make sure it is **off / unticked / disabled**:

NAME ON THE ROUTER	SEEN ON, FOR EXAMPLE
AP Isolation / Enable AP Isolation	TP-Link, Netgear, Tenda and many others
Set AP Isolated	ASUS
Client Isolation, **Wireless Isolation**, **Station Isolation**	various brands and ISP-supplied routers
SSID Isolation, **Privacy Separator**	ISP-supplied routers, Cisco/Linksys

If the router has two bands (2.4 GHz and 5 GHz), the setting may appear once for each: turn it off for both. Save, and restart the router if it asks.

3 Use the main network, not the guest network

A **guest network** keeps its devices apart by design, whatever the isolation setting says. Connect all laptops to the router's main Wi-Fi network. If the guest network is not needed, switch it off so that nobody joins it by mistake.

4 Give the Wi-Fi a clear name and a password

Rename the network to something unmistakable, for example `Optics-Lab`, so that it is not confused with phones and other routers nearby. Set the security to **WPA2-Personal** (or WPA2/WPA3) with a password and write the name and password on the board. Many routers come with an open network named like `TP-LINK_3BE4B2`: on an open network any phone in the corridor can join.

Both bands of one router (2.4 GHz and 5 GHz) are the same network: a laptop may use either, even when the router gives them different names.

5 Make sure there are enough addresses

The router hands each laptop an address (this is the **DHCP server**; leave it on). Under **Network → DHCP Server** or **LAN**, the address range must hold every laptop with some to spare. The usual range, for example `192.168.0.100` to `192.168.0.199`, holds 100 — enough for any class.

6 Optional: a fixed address for the teacher's laptop

Needed only when the students' laptops use *saved teachers' computers* (see section 3), because those remember the teacher's laptop by its address. Under **DHCP → Address Reservation** (or *Static Lease*, *Fixed IP*), pick the teacher's laptop from the list of connected devices and reserve its current address. It then gets the same address in every lesson.

3. The laptops

The teacher's laptop

- **Install with the teacher's installer** (`VirtualOpticsLab-Teacher-…-setup.exe`, 1.3.0 or later). During installation Windows asks once *"Do you want to allow this app to make changes?"* — choose **Yes**. That opens the classroom in Windows Firewall on every network, so nothing else needs setting.
- **Connect to the router's Wi-Fi.** Windows shows *No internet*: that is expected; stay connected.
- **Keep it from drifting away.** If the laptop knows a network with internet (a phone hotspot, the building's Wi-Fi), Windows may move to it during the lesson. Turn phone hotspots off in the room, or open that network in **Settings → Network & internet → Wi-Fi → Manage known networks** and untick *Connect automatically*.
- **Open the classroom.** Its screen shows the laptop's address, for example `192.168.0.104`. The first three numbers must match the router's (here `192.168.0`). An address such as `172.20.10.2` means the laptop is on a phone hotspot instead.

The students' laptops

- **Connect to the same Wi-Fi** as the teacher — the main network, not the guest one — and stay on it, as above.
- **Join an exam** on the start screen. The open classroom appears by itself, under the teacher's name.
- **If it does not appear: Settings → Find teachers on this network** asks every address on the network, which gets through routers that block the quick search. *Save this teacher* keeps it, or type the address shown on the teacher's screen.
- **Settings → Connection check** shows whether the laptop reaches the open classroom.
- The student laptops need no firewall change: they only send requests to the teacher's laptop.

> **Finding a laptop's own address.** In Windows: **Settings → Network & internet → Wi-Fi → (the network) → Properties**, line *IPv4 address*. Or press **Win + R**, type `cmd`, and in the window type `ipconfig`. Two laptops are on the same network when the first three numbers agree.

4. Before each exam — a five-minute check

- [ ] The router is on, and the Wi-Fi name on the board is the one the laptops see.
- [ ] Phone hotspots in the room are off.
- [ ] The teacher's laptop is on that Wi-Fi, and the classroom screen shows an address with the router's first three numbers.
- [ ] The teacher's laptop is plugged in (the application keeps it awake while a classroom is open).
- [ ] Two or three student laptops see the classroom under *Join an exam* before the whole class joins.
- [ ] The number of students joined matches the number in the room before pressing *Start the exam*.

5. When the classroom cannot be found

WHAT YOU SEE	LIKELY CAUSE	WHAT TO DO
No student finds the classroom; the teacher's screen shows an address	The laptops are on different networks	Compare the first three numbers of the teacher's address and a student's (see the note in section 3). If they differ, connect both to the router's main Wi-Fi.
Same network, still no student finds it, and typing the teacher's address does not work either	Windows Firewall on the teacher's laptop, or AP isolation on the router	**Firewall:** on the teacher's laptop set the Wi-Fi to *Private* (**Settings → Network & internet → Wi-Fi → the network → Network profile type**), or reinstall with the teacher's installer and accept the administrator prompt. If an antivirus program manages the firewall, allow *Virtual Optics Lab Teacher* there. **Router:** check step 2 again — AP isolation off on both bands, and not the guest network.
Typing the address works, but the classroom never appears by itself	The router passes ordinary traffic but drops the quick search	Use **Settings → Find teachers on this network** and *Save this teacher* on the students' laptops; give the teacher's laptop a fixed address (step 6).
Most students find it, a few do not	Those laptops joined another network (guest, a hotspot, a neighbouring router), or the router has reached its device limit	Check the Wi-Fi name on those laptops. If they are on the right one, disconnect a device that does not belong to the exam, or use a router that accepts more devices.
Saved teacher's computer stopped answering after a few days	The teacher's laptop got a new address from the router	Reserve its address on the router (step 6), then on the students' laptops *Find teachers on this network* once more.
A laptop drops off the Wi-Fi during the exam	Distance, or Windows moved to another known network	Nothing is lost: the student keeps working, the answers stay on the laptop and are sent when it reconnects. Reconnect it to the exam Wi-Fi.